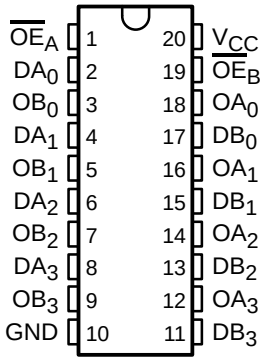
- **Function and Pinout Compatible With FCT and F Logic**
- **25-Ω Output Series Resistors Reduce Transmission-Line Reflection Noise**
- **TTL-Output-Level Version of Equivalent FCT Functions**
- **Edge-Rate Control Circuitry for Significantly Improved Noise Characteristics**
- **I_{off} Supports Partial-Power-Down Mode Operation**
- **ESD Protection Exceeds JESD 22**
  - **2000-V Human-Body Model (A114-A)**
  - **200-V Machine Model (A115-A)**
  - **1000-V Charged-Device Model (C101)**
- **Fully Compatible With TTL Input and Output Logic Levels**
- **12-mA Output Sink Current 15-mA Output Source Current**
- **3-State Outputs**

Q OR SO PACKAGE
(TOP VIEW)



description

The CY74FCT2244T is an octal buffer and line driver that includes on-chip 25-Ω terminating resistors at each of the outputs to minimize noise resulting from reflections or standing waves in high-performance applications. The on-chip resistors reduce overall board space and component count. Designed to be employed as a memory address driver, clock driver, and bus-oriented transmitter/receiver, this device provides speed and drive capabilities commensurate with its fastest bipolar logic counterparts, while reducing power dissipation. The input and output voltage levels allow direct interface with TTL, NMOS, and CMOS devices, without the need for external components.

This device is fully specified for partial-power-down applications using I_{off}. The I_{off} circuitry disables the outputs, preventing damaging current backflow through the device when it is powered down.

ORDERING INFORMATION

T_A	PACKAGE†		SPEED (ns)	ORDERABLE PART NUMBER	TOP-SIDE MARKING
–40°C to 85°C	QSOP – Q	Tape and reel	4.3	CY74FCT2244CTQCT	FCT2244C
	SOIC – SO	Tube	4.3	CY74FCT2244CTSOC	FCT2244C
		Tape and reel	4.3	CY74FCT2244CTSOCT	
	QSOP – Q	Tape and reel	4.6	CY74FCT2244ATQCT	FCT2244A
	SOIC – SO	Tube	4.6	CY74FCT2244ATSOC	FCT2244A
		Tape and reel	4.6	CY74FCT2244ATSOCT	
	QSOP – Q	Tape and reel	6.5	CY74FCT2244TQCT	FCT2244
	SOIC – SO	Tube	6.5	CY74FCT2244TSOC	FCT2244
		Tape and reel	6.5	CY74FCT2244TSOCT	

† Package drawings, standard packing quantities, thermal data, symbolization, and PCB design guidelines are available at www.ti.com/sc/package.

Please be aware that an important notice concerning availability, standard warranty, and use in critical applications of Texas Instruments semiconductor products and disclaimers thereto appears at the end of this data sheet.

PRODUCTION DATA information is current as of publication date. Products conform to specifications per the terms of Texas Instruments standard warranty. Production processing does not necessarily include testing of all parameters.

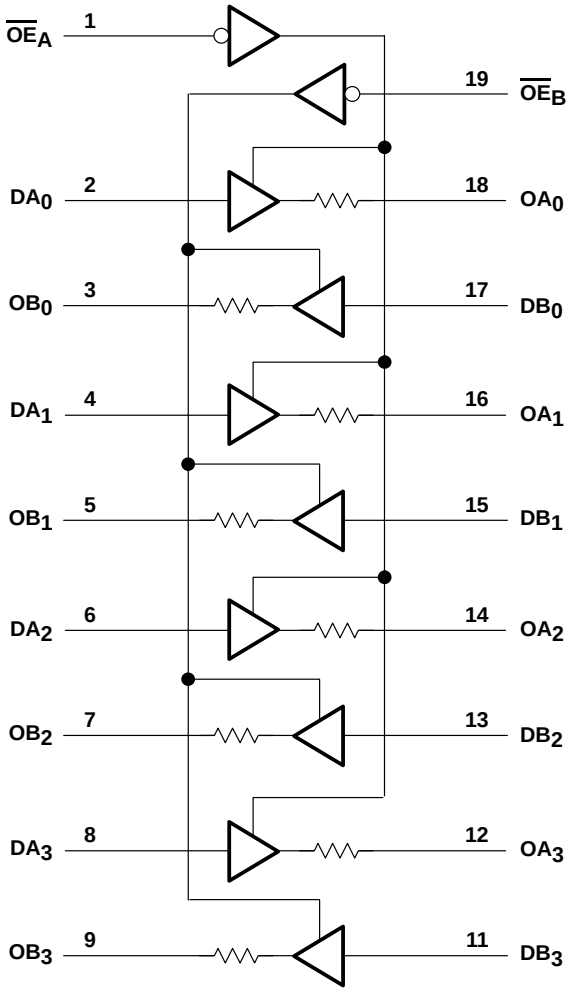
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FUNCTION TABLE

INPUTS			OUTPUT
$\overline{OE}_A$	$\overline{OE}_B$	D	O
L	L	L	L
L	L	H	H
H	H	X	Z

H = High logic level, L = Low logic level, X = Don't care, Z = High-impedance (off) state

logic diagram



absolute maximum ratings over operating free-air temperature range (unless otherwise noted)†

Supply voltage range to ground potential . . . . . –0.5 V to 7 V
DC input voltage range . . . . . –0.5 V to 7 V
DC output voltage range . . . . . –0.5 V to 7 V
DC output current (maximum sink current/pin) . . . . . 120 mA
Package thermal impedance, θ_{JA} (see Note 1): Q package . . . . . 68°C/W
SO package . . . . . 58°C/W
Ambient temperature range with power applied, T_A . . . . . –65°C to +135°C
Storage temperature range, T_{stg} . . . . . –65°C to +150°C

† Stresses beyond those listed under “absolute maximum ratings” may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under “recommended operating conditions” is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

NOTE 1: The package thermal impedance is calculated in accordance with JESD 51-7.

recommended operating conditions (see Note 2)

		MIN	NOM	MAX	UNIT
V_{CC}	Supply voltage	4.75	5	5.25	V
V_{IH}	High-level input voltage	2			V
V_{IL}	Low-level input voltage			0.8	V
I_{OH}	High-level output current			–15	mA
I_{OL}	Low-level output current			12	mA
T_A	Operating free-air temperature	–40		85	°C

NOTE 2: All unused inputs of the device must be held at V_{CC} or GND to ensure proper device operation.

electrical characteristics over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS			MIN	TYP†	MAX	UNIT
V_{IK}	V_{CC} = 4.75 V,	I_{IN} = –18 mA			–0.7	–1.2	V
V_{OH}	V_{CC} = 4.75 V,	I_{OH} = –15 mA		2.4	3.3		V
V_{OL}	V_{CC} = 4.75 V,	I_{OL} = 12 mA			0.3	0.55	V
R_{OUT}	V_{CC} = 4.75 V,	I_{OL} = 12 mA		20	25	40	Ω
V_{hys}	All inputs				0.2		V
I_I	V_{CC} = 5.25 V,	V_{IN} = V_{CC}				5	µA
I_{IH}	V_{CC} = 5.25 V,	V_{IN} = 2.7 V				±1	µA
I_{IL}	V_{CC} = 5.25 V,	V_{IN} = 0.5 V				±1	µA
I_{OZH}	V_{CC} = 5.25 V,	V_{OUT} = 2.7 V				10	µA
I_{OZL}	V_{CC} = 5.25 V,	V_{OUT} = 0.5 V				–10	µA
I_{OS}‡	V_{CC} = 5.25 V,	V_{OUT} = 0 V		–60	–120	–225	mA
I_{off}	V_{CC} = 0 V,	V_{OUT} = 4.5 V				±1	µA
I_{CC}	V_{CC} = 5.25 V,	V_{IN} ≤ 0.2 V,	V_{IN} ≥ V_{CC} – 0.2 V		0.1	0.2	mA
ΔI_{CC}	V_{CC} = 5.25 V, V_{IN} = 3.4 V§, f_1 = 0, Outputs open				0.5	2	mA
I_{CCD}¶	V_{CC} = 5.25 V, One input switching at 50% duty cycle, Outputs open, $\overline{OE}_A$ = $\overline{OE}_B$ = GND, V_{IN} ≤ 0.2 V or V_{IN} ≥ V_{CC} – 0.2 V				0.06	0.12	mA/ MHz
I_C#	V_{CC} = 5.25 V, Outputs open, $\overline{OE}_A$ = $\overline{OE}_B$ = GND	One bit switching at f_1 = 10 MHz at 50% duty cycle	V_{IN} ≤ 0.2 V or V_{IN} ≥ V_{CC} – 0.2 V		0.7	1.4	mA
			V_{IN} = 3.4 V or GND		1	2.4	
		Eight bits switching at f_1 = 2.5 MHz at 50% duty cycle	V_{IN} ≤ 0.2 V or V_{IN} ≥ V_{CC} – 0.2 V		1.3	2.6‖	
			V_{IN} = 3.4 V or GND		3.3	10.6‖	
C_i					5	10	pF
C_o					9	12	pF

† Typical values are at V_{CC} = 5 V, T_A = 25°C.

‡ Not more than one output should be shorted at a time. Duration of short should not exceed one second. The use of high-speed test apparatus and/or sample-and-hold techniques are preferable to minimize internal chip heating and more accurately reflect operational values. Otherwise, prolonged shorting of a high output can raise the chip temperature well above normal and cause invalid readings in other parametric tests. In any sequence of parameter tests, I_{OS} tests should be performed last.

§ Per TTL-driven input (V_{IN} = 3.4 V); all other inputs at V_{CC} or GND

¶ This parameter is derived for use in total power-supply calculations.

$I_C = I_{CC} + \Delta I_{CC} \times D_H \times N_T + I_{CCD} (f_0/2 + f_1 \times N_1)$

Where:

I_C = Total supply current
I_{CC} = Power-supply current with CMOS input levels
ΔI_{CC} = Power-supply current for a TTL high input (V_{IN} = 3.4 V)
D_H = Duty cycle for TTL inputs high
N_T = Number of TTL inputs at D_H
I_{CCD} = Dynamic current caused by an input transition pair (HLH or LHL)
f_0 = Clock frequency for registered devices, otherwise zero
f_1 = Input signal frequency
N_1 = Number of inputs changing at f_1

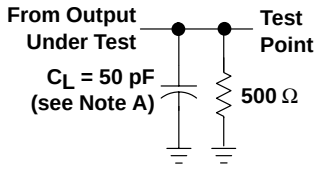
All currents are in milliamperes and all frequencies are in megahertz.

‖ Values for these conditions are examples of the I_{CC} formula.

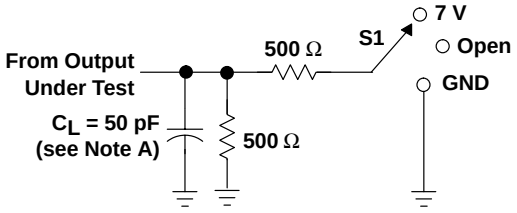
switching characteristics over operating free-air temperature range (see Figure 1)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	CY74FCT2244T		CY74FCT2244AT		CY74FCT2244CT		UNIT
			MIN	MAX	MIN	MAX	MIN	MAX	
t_{PLH}	D	O	1.5	6.5	1.5	4.6	1.5	4.1	ns
t_{PHL}			1.5	6.5	1.5	4.6	1.5	4.1	
t_{PZH}	$\overline{OE}$	O	1.5	8	1.5	6.2	1.5	5.8	ns
t_{PZL}			1.5	8	1.5	6.2	1.5	5.8	
t_{PHZ}	$\overline{OE}$	O	1.5	7	1.5	5.6	1.5	5.2	ns
t_{PLZ}			1.5	7	1.5	5.6	1.5	5.2	

PARAMETER MEASUREMENT INFORMATION

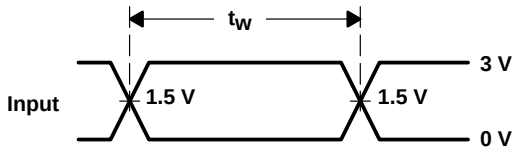


LOAD CIRCUIT FOR
TOTEM-POLE OUTPUTS

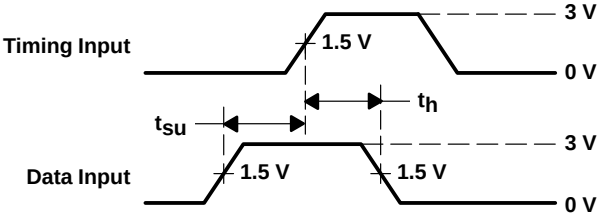


LOAD CIRCUIT FOR
3-STATE OUTPUTS

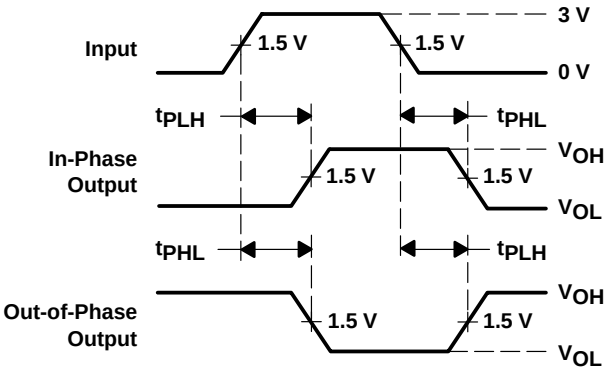
TEST	S1
t_{PLH}/t_{PHL}	Open
t_{PLZ}/t_{PZL}	7 V
t_{PHZ}/t_{PZH}	Open



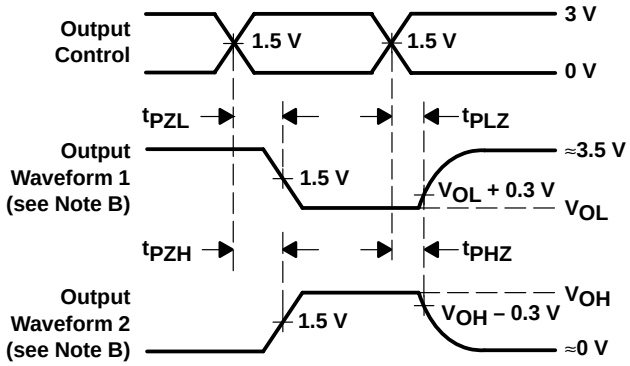
VOLTAGE WAVEFORMS
PULSE DURATION



VOLTAGE WAVEFORMS
SETUP AND HOLD TIMES



VOLTAGE WAVEFORMS
PROPAGATION DELAY TIMES
INVERTING AND NONINVERTING OUTPUTS



VOLTAGE WAVEFORMS
ENABLE AND DISABLE TIMES
LOW- AND HIGH-LEVEL ENABLING

NOTES: A. C_L includes probe and jig capacitance.
B. Waveform 1 is for an output with internal conditions such that the output is low except when disabled by the output control. Waveform 2 is for an output with internal conditions such that the output is high except when disabled by the output control.
C. The outputs are measured one at a time with one input transition per measurement.

Figure 1. Load Circuit and Voltage Waveforms

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
CY74FCT2244ATQCT	Active	Production	SSOP (DBQ) | 20	2500 | LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	FCT2244A
CY74FCT2244ATQCT.B	Active	Production	SSOP (DBQ) | 20	2500 | LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	FCT2244A
CY74FCT2244ATSOC	Active	Production	SOIC (DW) | 20	25 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	FCT2244A
CY74FCT2244ATSOC.B	Active	Production	SOIC (DW) | 20	25 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	FCT2244A
CY74FCT2244CTQCT	Active	Production	SSOP (DBQ) | 20	2500 | LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	FCT2244C
CY74FCT2244CTQCT.B	Active	Production	SSOP (DBQ) | 20	2500 | LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	FCT2244C
CY74FCT2244CTSOC	Active	Production	SOIC (DW) | 20	25 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	FCT2244C
CY74FCT2244CTSOC.B	Active	Production	SOIC (DW) | 20	25 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	FCT2244C
CY74FCT2244TQCT	Active	Production	SSOP (DBQ) | 20	2500 | LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	FCT2244
CY74FCT2244TQCT.B	Active	Production	SSOP (DBQ) | 20	2500 | LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	FCT2244
CY74FCT2244TSOC	Active	Production	SOIC (DW) | 20	25 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	FCT2244
CY74FCT2244TSOC.B	Active	Production	SOIC (DW) | 20	25 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	FCT2244
CY74FCT2244TSOCG4	Active	Production	SOIC (DW) | 20	25 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	FCT2244
CY74FCT2244TSOCG4.B	Active	Production	SOIC (DW) | 20	25 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	FCT2244

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

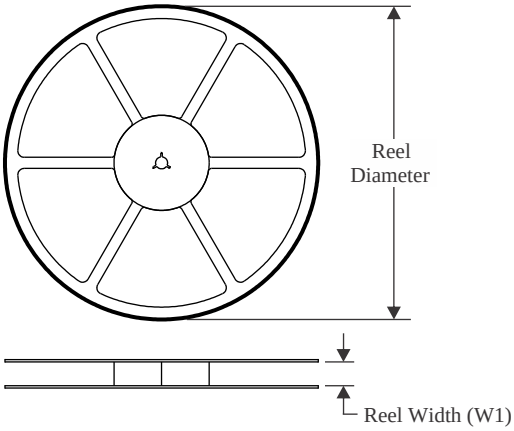
Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

Important Information and Disclaimer: The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

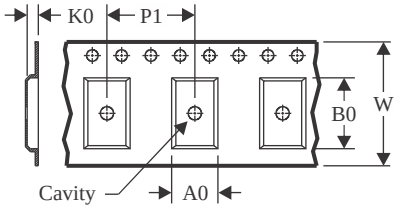
In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

TAPE AND REEL INFORMATION

REEL DIMENSIONS

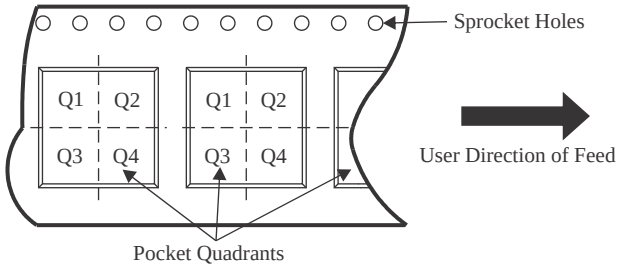


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

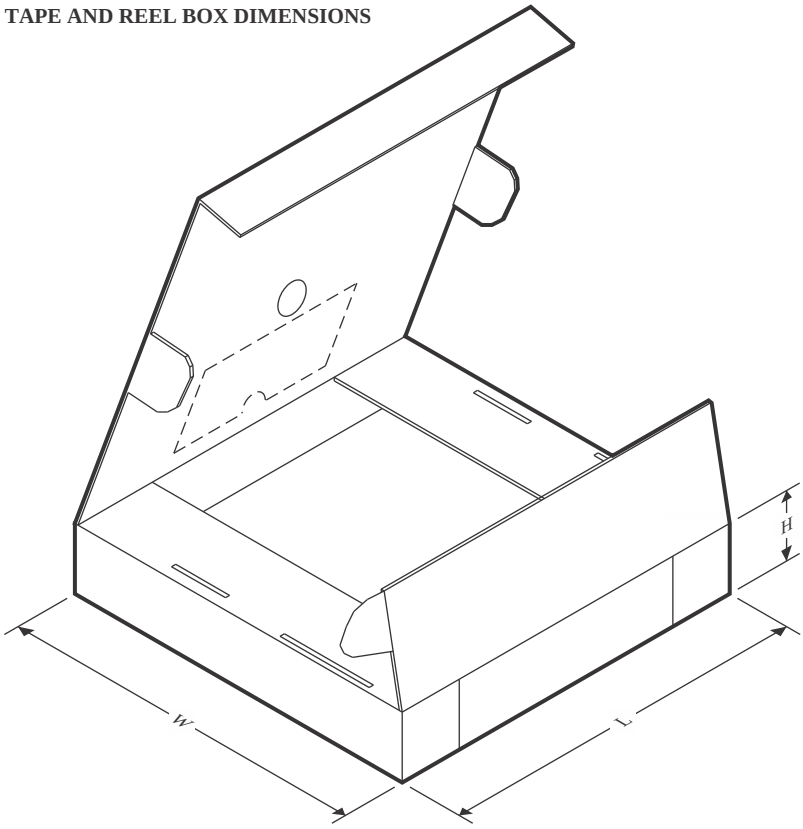
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
CY74FCT2244ATQCT	SSOP	DBQ	20	2500	330.0	16.4	6.5	9.0	2.1	8.0	16.0	Q1
CY74FCT2244CTQCT	SSOP	DBQ	20	2500	330.0	16.4	6.5	9.0	2.1	8.0	16.0	Q1
CY74FCT2244TQCT	SSOP	DBQ	20	2500	330.0	16.4	6.5	9.0	2.1	8.0	16.0	Q1

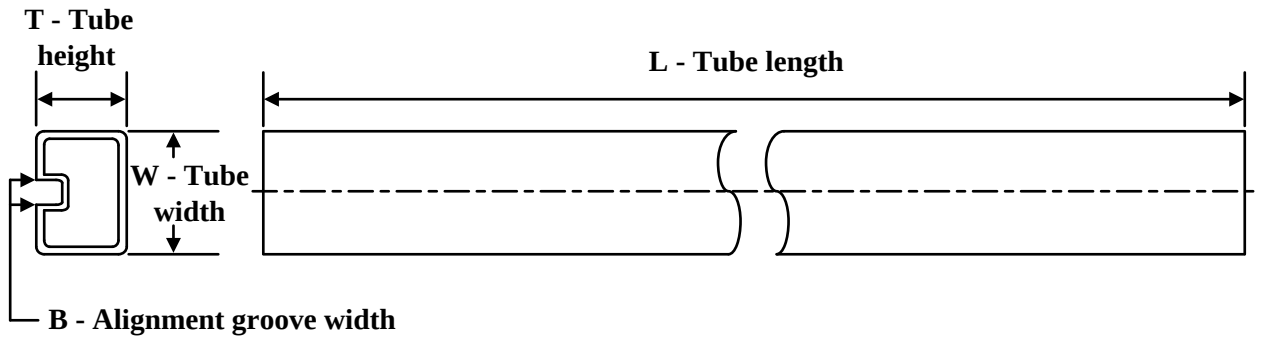
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
CY74FCT2244ATQCT	SSOP	DBQ	20	2500	353.0	353.0	32.0
CY74FCT2244CTQCT	SSOP	DBQ	20	2500	353.0	353.0	32.0
CY74FCT2244TQCT	SSOP	DBQ	20	2500	353.0	353.0	32.0

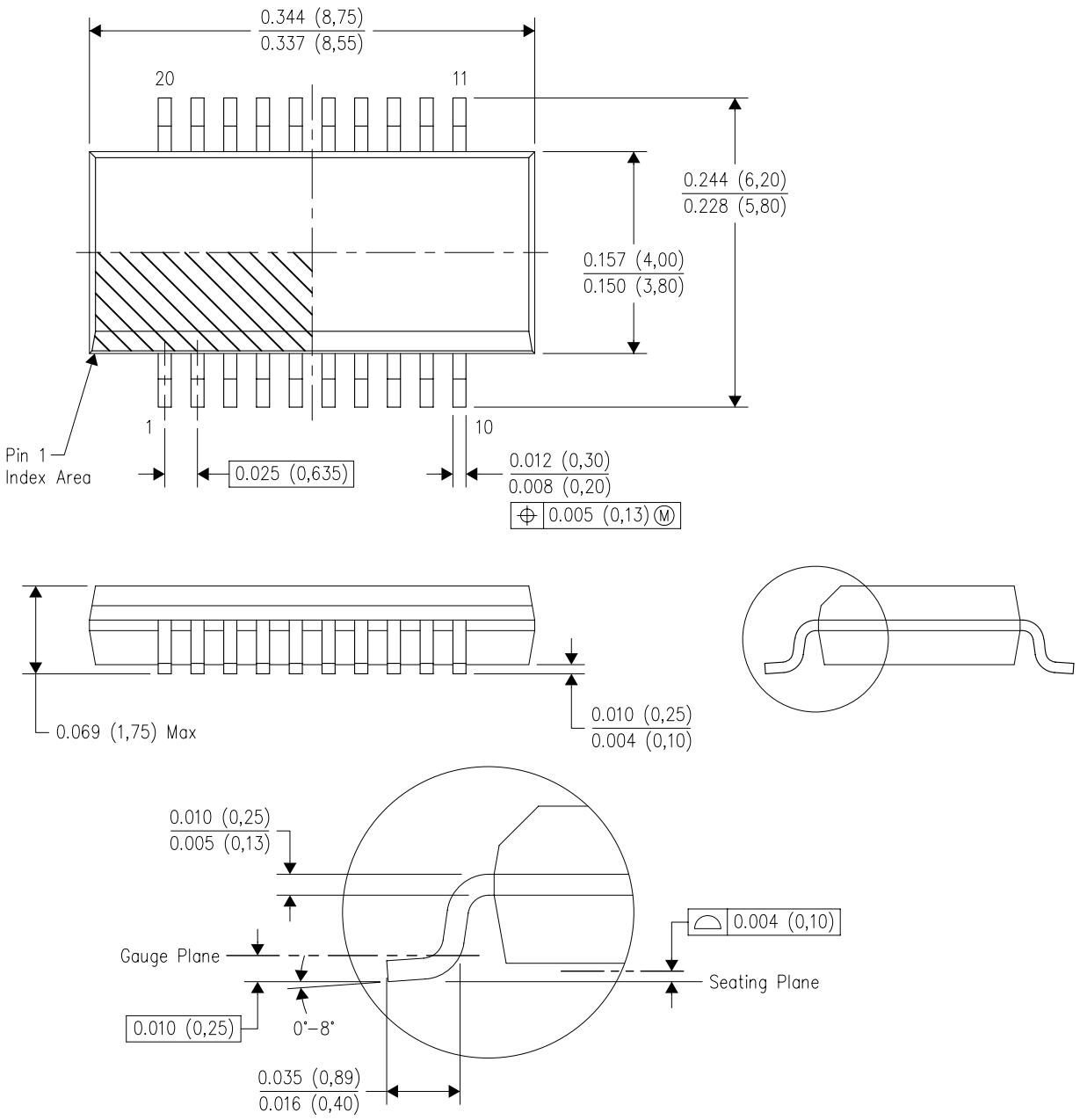
TUBE



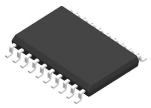
*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (µm)	B (mm)
CY74FCT2244ATSOC	DW	SOIC	20	25	507	12.83	5080	6.6
CY74FCT2244ATSOC.B	DW	SOIC	20	25	507	12.83	5080	6.6
CY74FCT2244CTSOC	DW	SOIC	20	25	507	12.83	5080	6.6
CY74FCT2244CTSOC.B	DW	SOIC	20	25	507	12.83	5080	6.6
CY74FCT2244TSOC	DW	SOIC	20	25	507	12.83	5080	6.6
CY74FCT2244TSOC.B	DW	SOIC	20	25	507	12.83	5080	6.6
CY74FCT2244TSOCG4	DW	SOIC	20	25	507	12.83	5080	6.6
CY74FCT2244TSOCG4.B	DW	SOIC	20	25	507	12.83	5080	6.6

DBQ (R–PDSO–G20) PLASTIC SMALL–OUTLINE PACKAGE



NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Body dimensions do not include mold flash or protrusion not to exceed 0.006 (0,15) per side.
D. Falls within JEDEC MO–137 variation AD.

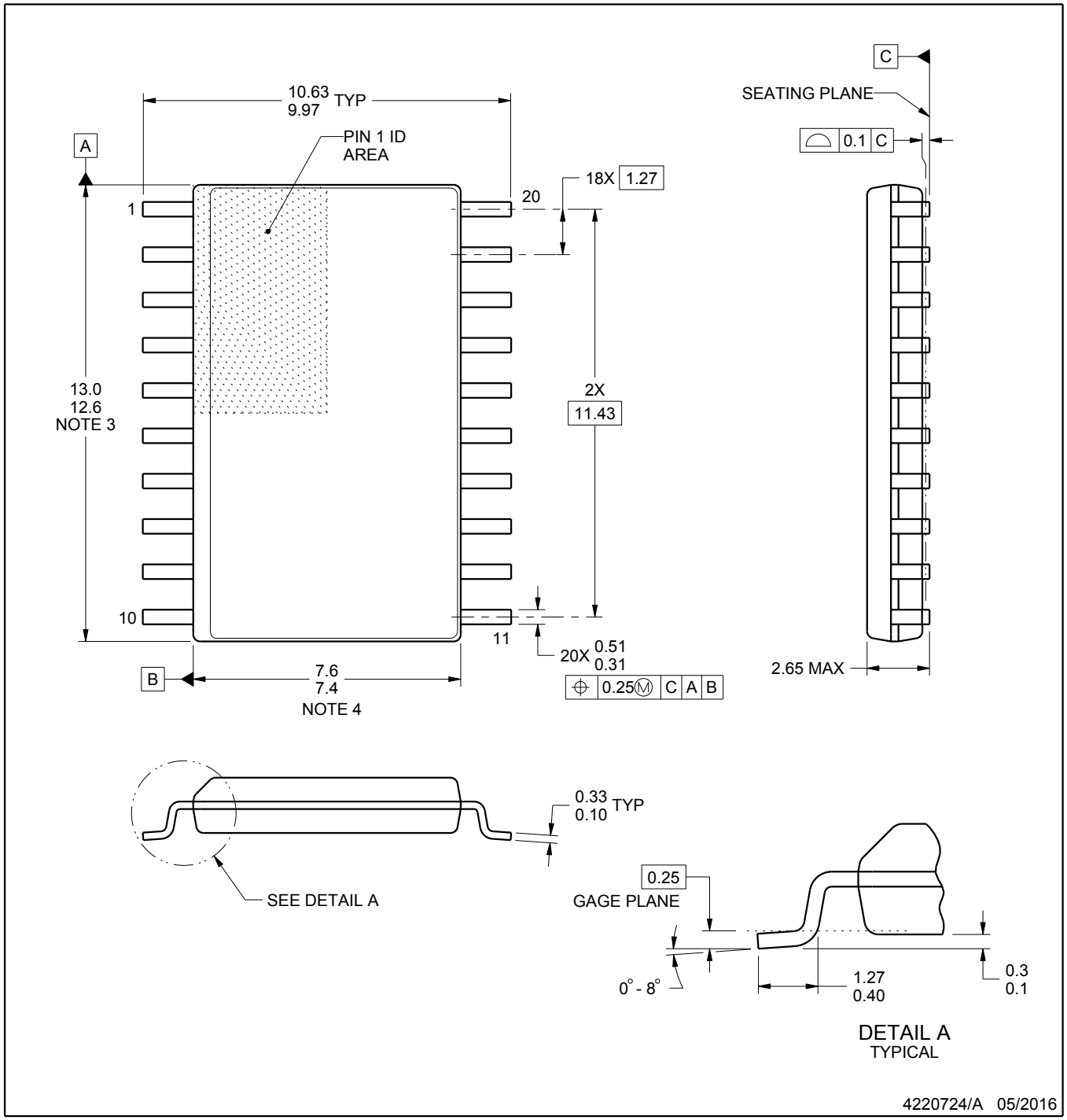


PACKAGE OUTLINE

DW0020A

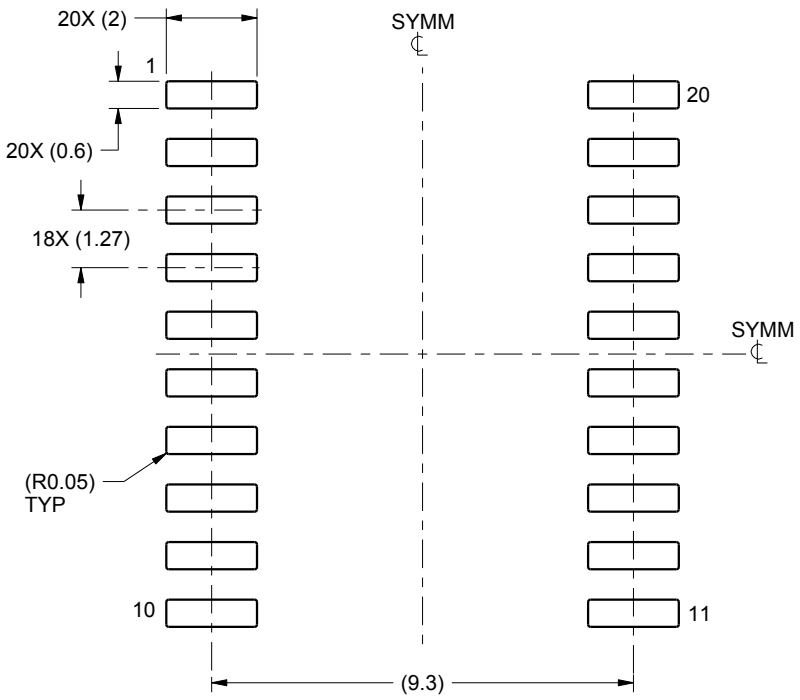
SOIC - 2.65 mm max height

SOIC

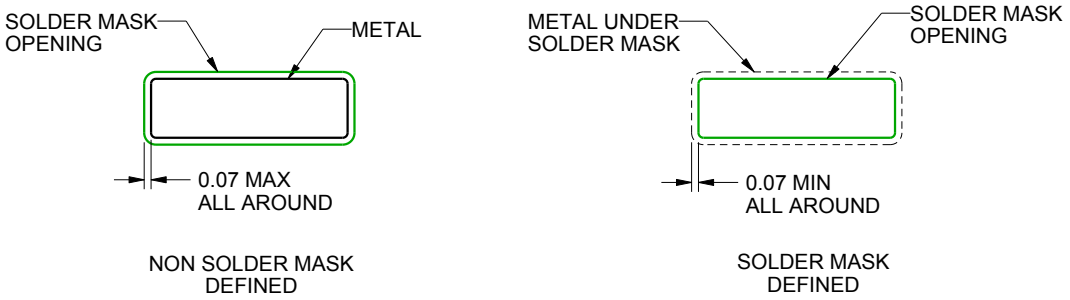


NOTES:

1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.43 mm per side.
5. Reference JEDEC registration MS-013.



LAND PATTERN EXAMPLE
SCALE:6X

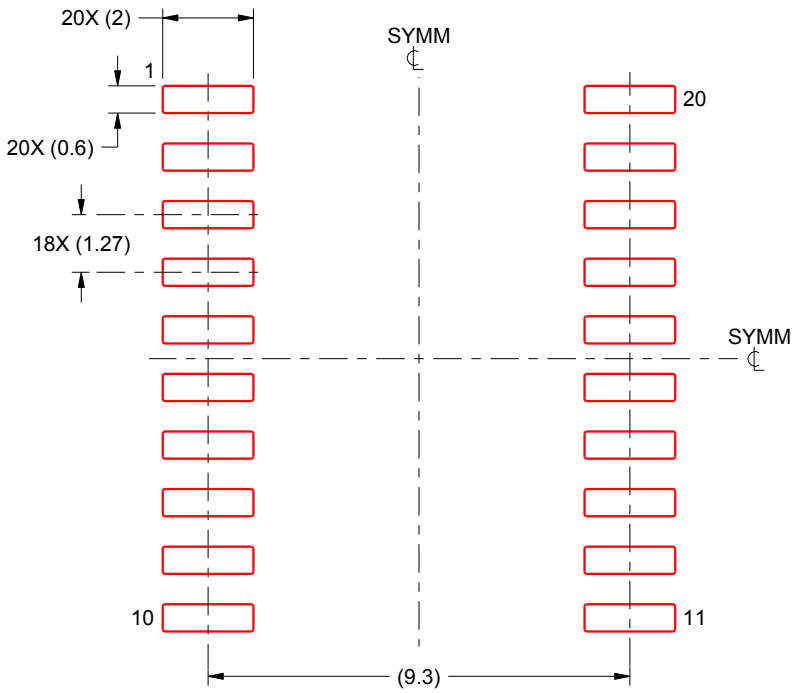


SOLDER MASK DETAILS

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NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:6X

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NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

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